

DUMBRAVENI, RomaniaWMO: **151890**Lat: **46.2278N**Lon: **24.5917E**Elev: **319**StdP: **97.55**Time Zone: **2.00 (EUE)**Period: **94-19**WBAN: **99999****Annual Heating, Humidification, and Ventilation Design Conditions**

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-15.4	-12.3	-17.0	0.9	-15.0	-13.8	1.2	-11.8	7.4	4.1	6.1	1.9	0.5	70	0.472

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
8	14.2	31.7	20.9	30.2	20.5	28.6	19.9	22.3	29.1	21.5	28.1	20.7	26.8	1.7	140

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
20.3	15.6	24.6	19.5	14.8	23.7	18.7	14.1	22.9	67.2	29.1	64.0	28.0	61.3	26.9	26.6

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 6.6	5.3	4.2	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A	N/A		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	10.1	-2.4	1.4	4.8	10.2	15.7	19.4	20.1	20.8	15.6	10.4	5.3	0.1
	DBStd	8.78	5.00	4.38	4.10	4.53	3.07	2.89	2.65	2.74	4.11	3.67	4.29	3.97
	HDD10.0	1349	386	242	164	54	2	0	0	0	4	41	150	306
	HDD18.3	3221	644	475	419	246	89	21	14	10	101	246	392	564
	CDD10.0	1403	0	0	4	59	180	282	312	333	171	54	8	0
	CDD18.3	233	0	0	0	0	9	53	68	85	17	0	0	0
	CDH23.3	2647	0	0	1	45	169	562	639	945	271	14	0	0
	CDH26.7	845	0	0	0	2	29	164	200	374	77	0	0	0
Wind	WSAvg	1.4	1.2	1.4	1.7	1.9	1.6	1.5	1.5	1.3	1.4	1.3	1.2	1.3
Precipitation	PrecAvg	643	26	24	35	53	80	101	85	62	61	50	35	33
	PrecMax	834	55	49	60	92	162	206	140	150	148	97	80	66
	PrecMin	454	6	3	4	12	30	18	28	12	5	6	0	5
	PrecStd	106	13	11	15	22	33	50	29	36	33	26	18	18
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	9.5	15.2	20.8	26.7	29.7	31.9	32.9	34.5	31.3	24.8	19.9	10.7
		MCWB	6.5	10.3	12.0	15.2	17.0	22.0	22.0	21.4	19.1	15.9	13.7	8.0
	2%	DB	6.6	12.6	17.5	24.7	27.4	30.4	30.7	32.1	28.7	23.1	16.4	8.0
		MCWB	4.8	8.3	10.2	14.9	17.5	21.1	21.5	20.5	18.6	14.9	12.1	6.6
	5%	DB	4.6	9.9	14.4	21.9	25.4	28.6	29.0	30.5	26.5	20.5	14.0	6.3
		MCWB	3.3	6.8	8.7	13.5	16.7	20.5	20.6	20.4	17.4	14.5	10.8	5.3
	10%	DB	2.9	7.4	11.9	19.3	23.2	26.7	27.0	28.7	23.9	17.5	11.8	4.7
		MCWB	2.0	5.2	7.3	12.2	15.7	19.6	19.9	20.0	17.0	13.0	9.3	4.0
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.9	10.9	12.5	16.5	20.7	23.6	23.5	23.1	21.1	18.1	13.9	8.4
		MCDB	8.8	15.5	20.2	24.6	26.7	29.8	29.4	30.9	27.5	22.2	18.3	10.6
	2%	WB	5.1	8.7	10.8	15.3	19.1	22.1	22.2	22.1	19.5	16.1	12.5	6.6
		MCDB	6.6	12.0	16.4	23.0	24.2	28.3	28.6	29.5	26.0	21.3	16.0	7.7
	5%	WB	3.5	7.1	9.3	14.3	17.8	21.1	21.3	21.2	18.4	14.9	11.5	5.3
		MCDB	4.4	9.4	13.9	21.1	23.0	27.1	27.3	28.2	25.0	19.4	13.9	6.1
	10%	WB	2.2	5.4	7.8	12.9	16.6	20.2	20.6	20.5	17.3	13.4	9.8	4.1
		MCDB	2.8	6.9	11.3	18.0	21.5	25.5	26.0	27.2	22.8	17.0	12.0	4.6
Mean Daily Temperature Range	5% DB	MDBR	7.8	9.7	11.4	13.0	12.2	12.7	12.7	14.2	13.1	12.7	9.7	6.0
		MCDBR	8.4	13.0	16.7	18.4	16.6	15.5	15.9	17.6	17.6	18.2	13.6	8.8
		MCWBR	6.4	9.1	9.9	9.1	7.6	6.7	7.1	7.0	7.6	10.5	9.0	6.9
	5% WB	MCDBR	8.1	13.0	15.8	17.0	14.5	14.0	14.2	16.1	16.0	16.1	12.6	8.2
	MCWBR	6.3	9.0	9.4	8.7	7.5	6.6	6.6	6.9	7.2	9.7	8.4	6.5	
Clear-Sky Solar Irradiance	taub	0.313	0.341	0.382	0.437	0.419	0.415	0.435	0.442	0.400	0.363	0.340	0.314	
	taud	2.330	2.285	2.236	2.110	2.227	2.271	2.222	2.199	2.282	2.375	2.387	2.357	
	Ebn at Noon	772	821	836	818	850	853	829	805	808	787	736	727	
	Edh at Noon	76	98	119	148	136	131	136	134	113	89	73	68	
All-Sky Solar Radiation	RadAvg	1.32	2.17	3.27	4.34	5.34	5.79	5.77	5.29	3.79	2.61	1.48	1.02	
	RadStd	0.16	0.25	0.37	0.44	0.49	0.50	0.49	0.47	0.41	0.32	0.18	0.12	